KAPASA MAKASA UNIVERSITY

AGRICULTURAL EXTENSION

DIFFUSION AND ADOPTION OF INNOVATIONS

1 DEFINITION OF DIFFUSION, ADOPTION AND INNOVATION

- **Diffusion** is the process by which an innovation is communicated through certain channels over time among the members of a social system (Rogers).
- **Innovation**: This is an idea, practice or object that is perceived as new by an individual or group.
- It doesn't matter if the idea is really new, it could have already been around for years.
- If the idea seems new to the individual or group, it is an innovation.
- **Adoption** means "accepting,"
- It differs from another commonly-used word adapt, which means "modify" or "change."

2 ELEMENTS OF DIFFUSION

Elements of diffusion influence the speed of diffusion of any innovation. These are:

a) The innovation itself (e.g., relative advantage, compatibility, complexity, trialability, observability).
   - Re-invention is the degree to which an innovation is changed or modified by a user in the process of its adoption and implementation.

b) The communication channels, i.e., the means by which messages get from one individual to another.

c) The time required to pass through the innovation dissemination process and the rate of adoption.
   - Time is involved in: Innovativeness, Innovation rate of adoption, Innovation Decision Process.
   - **(i) Innovativeness:** It is the degree to which an individual or other unit of adoption is relatively earlier in adopting new ideas than other members of a social system.
   - **(ii) Rate of Adoption:** It is the relative speed with which an innovation is adopted by members of a social system. Adoption is a decision to make full use of a new idea as the best course of action available.
   - **Rejection** is the decision not to adopt an innovation while **Discontinuance** is a decision to cease the use of an innovation

after adopting it earlier. Discontinuance, then, is essentially adoption of an innovation, followed by rejection. Discontinuance is of two types:

- Replacement: Replacement discontinuance is a decision to reject an idea in order to adopt a better idea that supersedes it.
- Disenchantment: Disenchantment discontinuance is a decision to reject an idea as a result of dissatisfaction with its performance.

- **(iii) The innovation-decision process**
- There are five levels that are passed through in order to adopt an innovation. These are:
  - Knowledge – acknowledging that the innovation exists
  - Persuasion – assuming a favourable attitude
  - Decision – choosing to go forward and try it
  - Implementation – trying it
  - Confirmation – permanent or long-term adoption of the innovation

d) The social system, including all the people or organizations that have joined in joint problem-solving to achieve a common goal or solution.

- It is a set of interrelated units that are engaged in joint problem solving to accomplish a common goal.
- **Components of a Social System**
  - Structure: It is patterned arrangement of the units.
  - Norms: These are the established behaviour patterns. Opinion leaders exhibit the norms.
- **Heterophily:** It is the degree to which pairs of individuals who interact are different in certain attributes, such as beliefs, education, social status and the like.
- **Homophily:** It is the degree to which pairs of individuals who interact are similar in certain attributes such as beliefs, education, social status and the like.

3 ATTRIBUTES OF INNOVATIONS

The following factors influence how soon they adopt an innovation.

a) Relative Advantage is the degree to which an innovation is perceived as better than the idea or practice it supersedes.

b) Compatibility is the degree to which an innovation is perceived as being consistent with the existing values, past experience, and needs of the potential adopters.
c) Complexity is the degree to which an innovation is perceived as difficult/ease to understand and use.
d) Trialability is the degree to which an innovation can be experimented with on a limited basis.
e) Observability is the degree to which the results of an innovation are visible to others.

Discussion

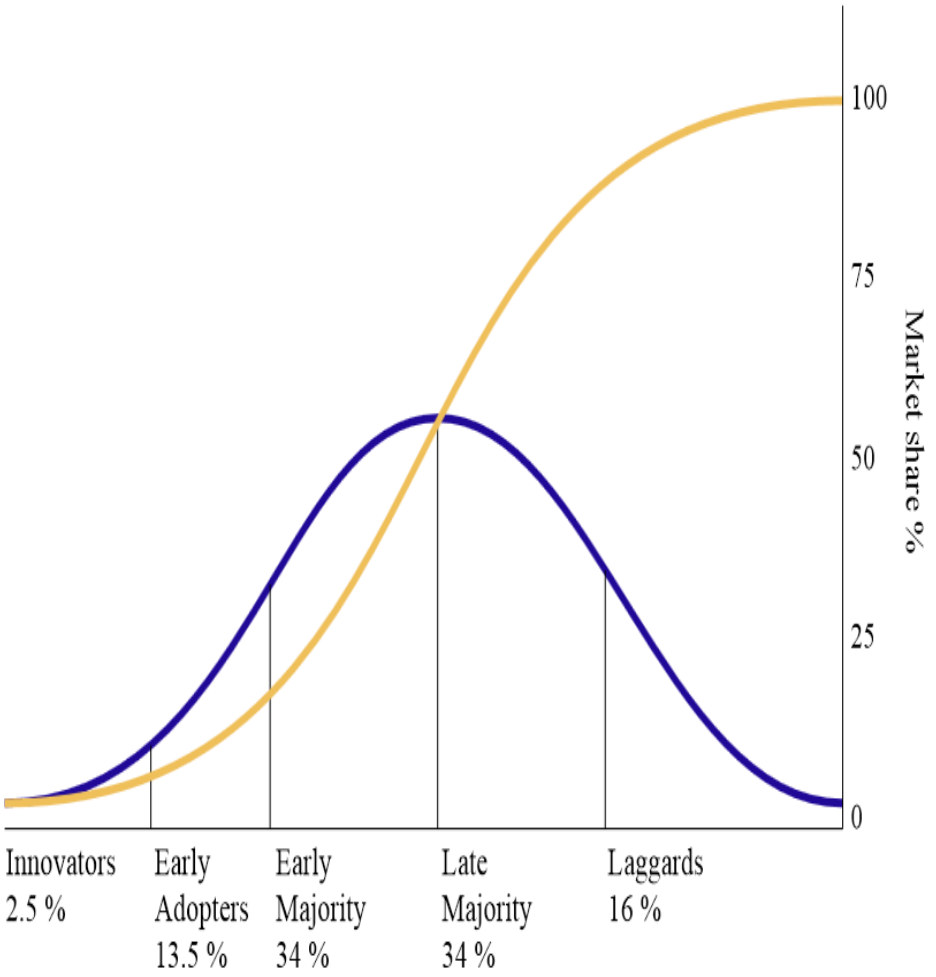
a) The farmer who said that fish pond is for organization x and that one is mine

b) A non-governmental organization is promoting improved fodder banks in Mulakupikwa agriculture camp. Improved fodder banks system/practice is being perceived as a new idea in this area, describe the general attributes of fodder banks system/practice that would promote its adoption by local people. [20 marks]

c)

4 INNOVATION ADOPTION CURVE

The figure below shows rate at which adoption takes place (yellow/gold curve) and categories of how individuals adopt an innovation (blue curve)



4.1 ADOPTER CATEGORIES.

- Individuals can be put into five groups,
- This grouping is based on time it takes the individuals to adopt a new innovation or behavior.
- The numbers of individuals in each of these groups forms a bell-shaped curve over time.
- These groups are the:
    - Innovators
    - Early adopters
    - Early majority
    - Late majority
    - Laggards
- NOTE: DON'T CALL THEM LAGGARDS OR WHATEVER.

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INNOVATORS

- These individuals are the first group to adopt an innovation and represent around 2.5% of the population.
- They are typically younger, they are “risk takers,” and they actively seek out new ideas.
- The innovators are highly curious and usually have adequate resources to absorb failure.
- They tend to have little or no influence in the decision making of the larger portion of the population, with respect to adopting an innovation.

EARLY ADOPTER

- These individuals are often the “opinion leaders” and represent about 13.5% of the population.
- They are, typically, older than the innovators, well-respected for their decision making, and often central in local communications networks.
- They tend to be adventurous but cautious and analytical in their adoption decisions.
- They have ample resources and low risk thresholds.
- The behaviors of this group can very easily influence the decisions of others in the population.

EARLY MAJORITY

- These individuals are highly connected to their peers, cautious, and take more time in making their decisions to adopt.
- They represent about 34% of a population.
- They tend to seek confirmation of the performance of an innovation and assurance through the number of people initially adopting an innovation.

LATE MAJORITY

- These individuals are much more skeptical of new ideas, are much more cautious, and have fewer resources to risk.
- They also make up around 34% of the population.
- They adopt an innovation out of economic necessity once they are assured of the results, as exhibited by the majority of the population who have already adopted the innovation.

LAGGARDS

- This this group represents 16% of the population.
- They are the last to adopt (or not) an innovation.
- They are typically the poorest and least socially connected.
- Being hesitant, they may end up adopting an innovation without having gone through all of the intermediate steps.

4.2 RATE OF ADOPTION

- In terms of the rate of adoption, an ‘S’ curve can be formed.
- Adoption of innovation starts out slowly, then accelerates once the innovators and early adopters have shown the advantage of adoption.
- The speed tails off, again, once the late majority have adopted, up until everyone in the “willing” population adopts.

5 FACTORS AFFECTING THE TRANSFER OF TECHNOLOGIES IN AGRICUTURAL DEVELOPMENT

- Social factors
    - Ethnocentrism,
    - Cultural lag and
    - Vested interest
- Nature of innovation
- The agriculture extension service deliverers